

Lecture 7

Complex Analysis II

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In today's lecture we continue our introduction to complex analysis by defining analytic complex functions, the complex derivative, and looking closer at rational complex functions.

Complex Derivatives

A complex function $f(s)$ is *analytic* in some domain $R \subset \mathbb{C}$ if the function is

- single valued, and
- has a finite complex derivative $f'(s) \equiv \frac{df}{ds}$ for all $s \in R$.

Let $f(s) = f(x + jy) = u(x, y) + jv(x, y)$. The function has a complex derivative if and only if

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

and

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

These are called the Cauchy-Riemann conditions. If these conditions are met then the complex derivative is given by

$$f'(s) = \frac{\partial u}{\partial x} + j \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - j \frac{\partial u}{\partial y}$$

If a complex function is analytic over the entire complex plane, i.e. $R = \mathbb{C}$, it is called an *entire* function.

If a complex function is not analytic at a point s_0 , but is analytic in the neighborhood of s_0 , then the point is called a *singularity* or a *singular point* of the function.

Examples

Example 1: Let $s = x + jy$ and define

$$f(s) = e^s = f(x + jy) = e^{x+jy} = e^x e^{jy}$$

Using Euler's relation

$$f(x + jy) = \underbrace{e^x \cos(y)}_{u(x,y)} + j \underbrace{e^x \sin(y)}_{v(x,y)}$$

Checking the Cauchy-Riemann conditions

$$\frac{\partial u}{\partial x} = e^x \cos(y)$$

$$\frac{\partial u}{\partial y} = -e^x \sin(y)$$

$$\frac{\partial v}{\partial x} = e^x \sin(y)$$

$$\frac{\partial v}{\partial y} = e^x \cos(y)$$

Thus $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = e^x \cos(y)$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = -e^x \sin(y)$ and the function is analytic everywhere and thus an entire function. The complex derivative is

$$f'(s) = e^x \cos(y) + j e^x \sin(y) = e^x e^{jy} = e^{x+jy} = e^s$$

just as if s was real and f a real function.

Example 2: Let $f(s) = \frac{1}{s}$, for $|s| > 0$, i.e. $s \neq 0$. Then

$$f(s) = f(x + jy) = \frac{1}{x + jy} = \frac{1}{x + jy} \cdot \frac{x - jy}{x - jy} = \frac{x - jy}{x^2 + y^2} = \underbrace{\frac{x}{x^2 + y^2}}_{u(x,y)} + j \underbrace{\frac{-y}{x^2 + y^2}}_{v(x,y)}$$

Checking the Cauchy-Riemann conditions

$$\frac{\partial u}{\partial x} = \frac{-x^2 + y^2}{(x^2 + y^2)^2}$$

$$\frac{\partial u}{\partial y} = \frac{-2xy}{(x^2 + y^2)^2}$$

$$\frac{\partial v}{\partial x} = \frac{2xy}{(x^2 + y^2)^2}$$

$$\frac{\partial v}{\partial y} = \frac{-x^2 + y^2}{(x^2 + y^2)^2}$$

we see that the function is analytic for $s \in \mathbb{C} - (0 + j0)$ and

$$f'(s) = \frac{-x^2 + y^2}{(x^2 + y^2)^2} + j \frac{2xy}{(x^2 + y^2)^2}$$

With some work you can show this is the same as

$$f'(s) = -\frac{1}{s^2}$$

again, just as if s was real and f a real function.

Example 3: Now for a counter-example. Consider

$$f(s) = s^* = f(x + jy) = x - jy$$

Thus $u(x, y) = x$ and $v(x, y) = -y$. Checking the Cauchy-Riemann conditions,

$$\frac{\partial u}{\partial x} = 1$$

$$\frac{\partial u}{\partial y} = 0$$

$$\frac{\partial v}{\partial x} = 0$$

$$\frac{\partial v}{\partial y} = -1$$

we see $\frac{\partial u}{\partial x} = 1 \neq \frac{\partial v}{\partial y} = -1$ thus the function is not analytic and no derivative exists.

Example 4: Show that $s_0 = -1$ is a singularity of an analytic complex function

$$f(s) = \frac{1}{s+1} = f(x + jy) = \frac{1}{(1+x) + jy}$$

Rearranging to get the real part and imaginary part

$$\frac{1}{(1+x)+jy} \frac{(1+x)-jy}{(1+x)-jy} = \underbrace{\frac{1+x}{(1+x)^2+y^2}}_{u(x,y)} + j \underbrace{\frac{-y}{(1+x)^2+y^2}}_{v(x,y)}$$

Checking the Cauchy-Riemann conditions,

$$\frac{\partial u}{\partial x} = \frac{-(1+x)^2+y^2}{[(1+x)^2+y^2]^2}$$

$$\frac{\partial u}{\partial y} = \frac{-2(1+x)y}{[(1+x)^2+y^2]^2}$$

$$\frac{\partial v}{\partial x} = \frac{2(1+x)y}{[(1+x)^2+y^2]^2}$$

$$\frac{\partial v}{\partial y} = \frac{-(1+x)^2+y^2}{[(1+x)^2+y^2]^2}$$

we see that $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$ except at $x = -1, y = 0$ where

$$\frac{-(1+x)^2+y^2}{[(1+x)^2+y^2]^2} = \frac{0}{0} = \text{undefined}$$

Thus the function is analytic except at $s_0 = -1 + j0 = -1$.

Rational Complex Functions

The previous example is a *rational function*, a ratio of polynomials in s . These will be very important later in the course. We will need to be very adept at doing manipulations of such functions. The general case can be written as

$$f(s) = \frac{\sum_{k=0}^M b_k s^k}{\sum_{k=0}^N a_k s^k} \equiv \frac{P(s)}{Q(s)}$$

- If $M < N$ the ratio is said to be *strictly proper*.
- If $M \leq N$ the ratio is *proper*.
- If $M > N$ the ratio is *improper*.

Example: $M = 2, N = 3$

$$f(s) = \frac{b_0 + b_1s + b_2s^2}{a_0 + a_1s + a_2s^2 + a_3s^3}$$

is a strictly proper rational function.

The roots of the denominator polynomial $Q(s)$ are the singularities of f .

Partial Fraction Expansion

We will often be interested in writing rational functions as a *partial fraction expansion* of the factors of the denominator. A ratio of polynomials that is improper can be written as the sum of a polynomial and a strictly proper rational function. Example

$$f(s) = \frac{2s^3 + 9s^2 + 11s + 2}{s^2 + 4s + 3} = (2s + 1) + \frac{s - 1}{s^2 + 4s + 3}$$

Once a rational function is proper we can expand it in terms of the factors of $Q(s)$. Example

$$f(s) = \frac{1}{s^2 + 3s + 2} = \frac{1}{(s + 1)(s + 2)} = \frac{A}{s + 1} + \frac{B}{s + 2}$$

To find A and B we can "clear" the fractions

$$\frac{A}{s + 1} + \frac{B}{s + 2} = \frac{1}{(s + 1)(s + 2)} = \frac{A(s + 2) + B(s + 1)}{(s + 1)(s + 2)}$$

which gives the equation for the numerator

$$A(s + 2) + B(s + 1) = 1 + 0s$$

which requires $A = 1$ and $B = -1$, so that

$$f(s) = \frac{1}{s + 1} + \frac{-1}{s + 2}$$

Clearing fractions can be tedious with higher order systems. A shortcut is to use the Heaviside "cover-up" method, or finding the *residues*.

Case #1: non-repeated singularities

$$f(s) = \frac{P(s)}{(s - \alpha_1)(s - \alpha_2) \cdots (s - \alpha_N)}$$

where α_i are the distinct singularities (roots of $Q(s)$). Then

$$f(s) = \frac{K_1}{s - \alpha_1} + \frac{K_1}{s - \alpha_1} + \cdots + \frac{K_N}{s - \alpha_N}$$

To find the K_i values we multiply through by that term $(s - \alpha_i)$ and evaluate the result when $s = \alpha_i$

$$K_i = (s - \alpha_i)f(s)|_{s=\alpha_i}$$

Example:

$$f(s) = \frac{s^2 + 2s + 7}{(s + 1)(s - 3)(s + 5)} = \frac{K_1}{s + 1} + \frac{K_2}{s - 3} + \frac{K_3}{s + 5}$$

We note that $M = 2$ and $N = 3$ so that the function is strictly proper. To find the constants

$$K_1 = \frac{s^2 + 2s + 7}{(s - 3)(s + 5)} \Big|_{s=-1} = -\frac{3}{8}$$

$$K_2 = \frac{s^2 + 2s + 7}{(s + 1)(s + 5)} \Big|_{s=3} = \frac{11}{16}$$

$$K_3 = \frac{s^2 + 2s + 7}{(s + 1)(s - 3)} \Big|_{s=-5} = \frac{11}{16}$$

This works even if the roots are complex. For example:

$$f(s) = \frac{1}{(s + 1 + j)(s + 1 - j)} = \frac{K_1}{s + 1 + j} + \frac{K_2}{s + 1 - j}$$

$$K_1 = \frac{1}{s + 1 - j} \Big|_{s=-1-j} = -\frac{1}{2j}$$

$$K_2 = \frac{1}{s + 1 + j} \Big|_{s=-1+j} = \frac{1}{2j}$$

Case #2: However we will often want to avoid working with complex roots directly. Another approach is to combine them into a quadratic term. Here is an example:

$$f(s) = \frac{s + 2}{s^3 + 3s^2 + 4s + 2} = \frac{s + 2}{(s + 1)(s + 1 + j)(s + 1 - j)} = \frac{s + 2}{(s + 1)(s^2 + 2s + 2)} = \frac{A}{s + 1} + \frac{Bs + C}{s^2 + 2s + 2}$$

To find A we use the same residue method

$$A = \left. \frac{s+2}{s^2+2s+2} \right|_{s=-1} = 1$$

To find B and C we can clear fractions or use another shortcut. To find C , let $s = 0$. We get

$$\frac{A}{1} + \frac{C}{2} = 1$$

which implies that $C = 0$. To find B multiply through by s and let $s \rightarrow \infty$.

$$\frac{As}{s+1} + \frac{Bs^2+Cs}{s^2+2s+2} = \frac{s^2+2s}{s^3+3s^2+4s+2}$$

In the first term divide top and bottom by s , in the second term by s^2 , and on the right-hand side by s^3 :

$$\frac{A}{1+s^{-1}} + \frac{B+Cs^{-1}}{1+2s^{-1}+2s^{-2}} = \frac{s^{-1}+2s^{-2}}{1+3s^{-1}+4s^{-2}+2s^{-3}}$$

Now let $s \rightarrow \infty$. We get

$$\frac{A}{1} + \frac{B}{1} = \frac{0}{1} = 0$$

which implies $B = -1$.

Or you can just clear the fractions and substitute for A and C .

$$As^2 + 2As + 2A + Bs^2 + Cs + Bs + C = s + 2 \quad (1)$$

$$(A+B)s^2 + (2A+B+C)s + 2A+C = s+2 \quad (2)$$

$$(3)$$

We only need one equation, so let's use $2A+B+C=1$. Solving for B and substituting for A, C gives $B = -1$ as above.

Thus the final result is

$$f(s) = \frac{1}{s+1} + \frac{-s}{s^2+2s+2}$$

Case # 3: One other complication is when we have repeated roots (a root λ , repeated $r > 1$ times).

$$f(s) \frac{P(s)}{(s-\lambda)^r (s-\alpha_1)(s-\alpha_2) \cdots (s-\alpha_{N-r})}$$

To handle this case we expand the ratio as

$$f(s) = \frac{a_0}{(s-\lambda)^r} + \frac{a_1}{(s-\lambda)^{r-1}} + \cdots + \frac{a_{r-1}}{s-\lambda} + \frac{K_1}{s-\alpha_1} + \frac{K_2}{s-\alpha_2} + \cdots + \frac{K_{N-r}}{s-\alpha_{N-r}}$$

The K_i values are determined using the same residue method as before. The a_i values are found using a variation on the residue method:

$$a_i = \frac{1}{i!} \frac{d^i}{ds^i} [(s-\lambda)^r f(s)] \Big|_{s=\lambda}$$

That is rather complicated so lets finish up with an example. Find the partial fraction expansion of

$$f(s) = \frac{s+3}{(s+1)^2(s+2)}$$

Using the pattern above we expand the terms as

$$f(s) = \frac{K_1}{s+2} + \frac{a_0}{(s+1)^2} + \frac{a_1}{s+1}$$

then find the residues

$$K_1 = \frac{s+3}{(s+1)^2} \Big|_{s=-2} = 1$$

$$a_0 = \frac{1}{0!} \frac{d^0}{ds^0} \left[\frac{s+3}{s+2} \right] \Big|_{s=-1} = 2$$

$$a_1 = \frac{1}{1!} \frac{d}{ds} \left[\frac{s+3}{s+2} \right] \Big|_{s=-1} = \frac{(s+2)(1) - (s+3)(1)}{(s+2)^2} \Big|_{s=-1} = -1$$

Thus the result is

$$f(s) = \frac{1}{s+2} + \frac{2}{(s+1)^2} + \frac{-1}{s+1}$$

As you might expect there are computational tools that do this job for you.

- Mathematica: `Apart` function
- Matlab (using symbolic toolbox) and Maxima: `partfrac` function

You can use this to check work and help on homework, but you will need to be able to do this by hand on exams.